Lab 2 - Requirement and Specification (SRS)

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1 Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to outline the requirements of the BevOS prototype being developed. This document explains the main functions of the system, who will be using it, and what is expected with the finished prototype. It also identifies limitations and dependencies that could affect development.

This SRS will be a reference for team silver to use throughout the project development. By documenting the requirements, the team can make sure everyone has a clear understanding of what needs to be developed and how the parts will work together.

1.2 Scope

BevOS is intended to centralize information associated with beverage manufacturing workflows rather than replace every system used. The platform supports product requests, supplier quotes, approvals, production tracking, logistics coordination, document storage, and communication. Authorized users can monitor current workflow information through a shared platform while maintaining responsibilities associated with their individual roles. The prototype focuses on demonstrating how these capabilities can reduce separated communication and improve visibility.

1.3 Definitions, Acronyms, and Abbreviations

The following definitions are based on the glossary established by Team Silver in Lab 1.

- **Approval Workflow** - A process used to review and approve operational tasks before they move to the next stage.
- **BevOS** - A centralized workflow-management platform designed for beverage manufacturing operations.
- **Client** - An organization or individual purchasing services or products from another participant in the manufacturing process.
- **Co-Packer** - A third-party company that manufactures and packages beverages for another company.
- **Core Services** - The shared BevOS system software layer that processes requests, applies system rules and coordination functional modules, and manages data exchange.
- **Dashboard** - A visual interface displaying operational information and performance metrics.
- **Distributor** - An organization responsible for delivering products from manufacturers to retailers or customers.
- **ERP (Enterprise Resource Planning)** - Software that integrates core business processes into a unified system.
- **Logistics** - The planning and coordination of transportation and delivery activities.
- **MFCD (Major Functional Components Diagram)** - A diagram illustrating the primary hardware and software components of the system.

- **Nice Package Company** - The beverage-industry client and domain partner for whom BevOS is being developed.
- **PostgreSQL** - An open-source relational database management system used to store application data.
- **Role-Based Access Control** - A security method that grants system permissions according to the user's assigned role and responsibilities.
- **Single Source of Truth** - A trusted, centralized set of current information used consistently by authorized stakeholders.
- **Stakeholder** - A person or organization that participates in or is affected by a beverage-manufacturing workflow.
- **Workflow** - A sequence of business activities completed to achieve a specific objective.

1.4 References

- Grand View Research. (2024). *Beverage Packaging Market Size, Share & Trends Analysis Report.*
- Kennedy, T. (2026). *CS410 Professional Workforce Development.*
- Wilson, T. (2026). *BevOS Presentation / Nice Package Co. Guest Presentation.*
- Zimmerman, S. (2026) *How the food and beverage industry is rethinking manufacturing investments. Food Dive.*

1.5 Overview

Section 2 provides an overall description of BevOS, including the systems structure, main features, intended users, constraints, and dependencies. This section explains what the prototype is expected to be able to do without going into to much detail on the specific requirements of each component.

Section 3 contains the specific requirements for the BevOS prototype. These requirements explain how the system should function, the expected performance, and other requirements needed for development.

2 Overall Description

2.1 Product Perspective

BevOS centralizes requests, approvals, documents, communication, production, and logistics information into one system with a purpose to reduce the need for separate applications.

Product request management allows authorized users to create, review, and track requests throughout the workflow, quote management organizes supplier quotes and keeps them connected to their related requests, reducing unnecessary follow-ups.

Approval Tracking manages artwork, purchase orders, and other approvals while recording their status and history. This helps users identify delays and keep workflow moving efficiently.

2.2 Product Functions

BevOS will provide several functions that support the management of beverage manufacturing workflows. The prototype will focus on demonstrating how these functions interact and how information can be shared between components.

2.2.1 Workflow and Request Management

BevOS will provide a centralized location for managing product requests and related business activities. Authorized users will be able to access request information and monitor its progress throughout the workflow.

2.2.2 Quote and Approval Management

The system will support the organization of supplier quotes and approval requests. Quote information will be associated with the appropriate business request so users can review the relevant details when needed. Approval tracking will allow users to identify the status of pending or completed approvals.

2.2.3 Production Management

BevOS will support the tracking and production information, including schedules, progress updates, and status changes. Production records will remain connected to the appropriate business workflows so users can monitor manufacturing activities.

2.2.4 Logistics Coordination

The logistics component will provide information related to shipping and delivery activities. Users will be able to monitor recorded shipment statuses and access information associated with the transportation of products.

2.2.5 Document Management

BevOS will provide a centralized location for organizing documents related to business operations. These documents may include artwork, purchase orders, production records, and shipping information.

2.2.6 Communication and Reporting

The communication component is intended to keep important updates connected to their related business activities. This will help prevent information from becoming separated from the requests or tasks it belongs to.

Reporting and dashboard features will provide users with a way to view operational information in an organized format. These features are intended to make it easier to monitor business activity and identify areas that need attention.

2.2.7 User Interface

BevOS will provide a web interface that allows users to navigate between the different parts of the application. The interface will include navigation menus, dashboards, and pages for accessing business information.

2.3 User Characteristics

BevOS is intended for several types of users involved in beverage manufacturing. Each user has different responsibilities within the production process, so the information they need from the system will vary depending on role.

2.3.1 Brands and Clients

Brands and clients are responsible for requesting products and monitoring their progress. They may need to submit requests, review approval statuses, and check production information.

2.3.2 Distributors

Distributors help coordinate the movement and availability of products. They may use BevOS to review business requests, track product information, and monitor delivery activity.

2.3.3 Suppliers

Suppliers provide materials, packaging, and other services needed for beverage production. They may use the system to review requests, provide quotes, and access information related to their responsibilities.

2.3.4 Production and Logistics Personnel

Production and logistics personnel are responsible for managing manufacturing activities, schedules, shipments, and deliveries. They will need access to current operational information so they can coordinate tasks and identify potential delays.

2.3.5 Management and Administrative Users

Management and administrative users are responsible for overseeing different areas of business operations. They may use BevOS to review reports, monitor workflow progress, and manage information within the system.

2.4 Constraints

N/A

2.5 Assumptions and Dependencies

N/A